Hydrogen Outlook

1. Production routes

Most hydrogen produced today is grey hydrogen, made from natural gas through steam methane reforming, which releases carbon dioxide.

Blue hydrogen uses the same process but captures and stores most of the carbon dioxide emissions.

Green hydrogen is produced by electrolysis, which splits water into hydrogen and oxygen using electricity from renewable sources.

2. Electrolysis efficiency

Current commercial electrolysers convert roughly 60 to 70 percent of the input electricity into the chemical energy of hydrogen.

Proton exchange membrane (PEM) electrolysers respond quickly to changes in power input, which makes them well suited to variable wind and solar supply.

3. Storage and use

Hydrogen can be stored for long periods in salt caverns, which makes it a candidate for seasonal energy storage.

Converting electricity to hydrogen and back to electricity has a round-trip efficiency of only about 30 to 40 percent, far below lithium-ion batteries.

Hard-to-electrify sectors such as steelmaking, fertiliser production and long-haul shipping are considered the most promising uses of green hydrogen.